Recording L01-R01
Marginal and conditional Gaussian laws, worked carefully

Block selection and an independent Gaussian residual.
$p(x_1), \quad p(x_1 \mid x_2)$

Probabilistic Machine Learning 2027-1

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YOUR NOTES

Recording L01-R01 — Gaussian conditioning

From the lecture to the two laws we will work through

From the lecture L01 S47–S49 · Gaussian marginal and conditional laws

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim \mathcal{N}(\mu, \Sigma)$$

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \succ 0$$

To work through here

$$p(x_1) = \int p(x_1, x_2)\, \mathrm{d}x_2$$

Select a variable block.

$$p(x_1 \mid x_2) = \frac{p(x_1, x_2)}{p(x_2)}$$

Construct an independent Gaussian residual.

- Work through the Gaussian laws established in the prepared lecture.

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STUDY NOTE

The lecture supplied the marginal and conditional Gaussian formulas. Here we will derive the marginal by selecting a block, then derive the conditional law by constructing a residual independent of the observed block.

YOUR NOTES

Recording L01-R01 — Gaussian conditioning

Aside: A linear map carries the Gaussian mean and covariance

Carried forward S02 · joint Gaussian block model; fixed T

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \right)$$

Mean and covariance

$$E[TX] = T\mu, \quad TX - E[TX] = T(X - \mu)$$

$$\begin{aligned} \mathrm{Cov}(TX) &= E[(TX - E[TX])(TX - E[TX])^T] \\ &= TE[(X - \mu)(X - \mu)^T]T^T \\ &= T\Sigma T^T \end{aligned}$$

Gaussian projection

Every linear projection of a Gaussian is Gaussian.

$$TX \sim \mathcal{N}(T\mu, T\Sigma T^T)$$

- Block selection is rectangular, so no inverse of T is needed.

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STUDY NOTE

A linear image of a Gaussian remains Gaussian, with mean and covariance transformed by the same matrix. This rule is enough to derive a marginal law because selecting a block is a linear map, even when its matrix is rectangular.

YOUR NOTES

Recording L01-R01 — Gaussian conditioning

Selecting a variable block is a linear transformation

Select the first block

$$P_1 = \begin{pmatrix} I_{d_1} & 0 \end{pmatrix}$$

$$P_1 X = \begin{pmatrix} I_{d_1} & 0 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = X_1$$

$$X_1 \sim \mathcal{N}(P_1 \mu, P_1 \Sigma P_1^T)$$

Select the second block

$$P_2 = \begin{pmatrix} 0 & I_{d_2} \end{pmatrix}$$

$$P_2 X = \begin{pmatrix} 0 & I_{d_2} \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = X_2$$

$$X_2 \sim \mathcal{N}(P_2 \mu, P_2 \Sigma P_2^T)$$

- Each selected block is Gaussian.
- The selector products determine its parameters.

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STUDY NOTE

Each selector keeps one component of the joint vector. Applying the affine Gaussian rule to these selectors establishes that both blocks are Gaussian and reduces the marginal calculation to matrix products.

YOUR NOTES

Recording L01-R01 — Gaussian conditioning

Each marginal keeps its matching mean and covariance blocks

Carried forward S03 · affine Gaussian transformation

$X \sim \mathcal{N}(\mu, \Sigma) \Rightarrow AX + b \sim \mathcal{N}(A\mu + b, A\Sigma A^T)$

$$X_i \sim \mathcal{N}(P_i\mu, P_i\Sigma P_i^T), \quad i = 1, 2$$

First block

$$P_1\mu = \begin{pmatrix} I_{d_1} & 0 \end{pmatrix}\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} = \mu_1$$

$$P_1\Sigma P_1^T = \begin{pmatrix} I_{d_1} & 0 \end{pmatrix}\begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}\begin{pmatrix} I_{d_1} & 0 \end{pmatrix}^T$$

$$= \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \end{pmatrix}\begin{pmatrix} I_{d_1} & 0 \end{pmatrix}^T = \Sigma_{11}$$

$$p(x_1) = \mathcal{N}(x_1 \mid \mu_1, \Sigma_{11})$$

Second block

$$P_2\mu = \begin{pmatrix} 0 & I_{d_2} \end{pmatrix}\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} = \mu_2$$

$$P_2\Sigma P_2^T = \begin{pmatrix} 0 & I_{d_2} \end{pmatrix}\begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}\begin{pmatrix} 0 & I_{d_2} \end{pmatrix}^T$$

$$= \begin{pmatrix} \Sigma_{21} & \Sigma_{22} \end{pmatrix}\begin{pmatrix} 0 & I_{d_2} \end{pmatrix}^T = \Sigma_{22}$$

$$p(x_2) = \mathcal{N}(x_2 \mid \mu_2, \Sigma_{22})$$

- Both marginal laws follow from selection without density integration.

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STUDY NOTE

Multiplying the selectors into the joint mean and covariance picks out the corresponding diagonal blocks. Thus the marginal law of either block follows without integrating the joint density.

YOUR NOTES

Recording L01-R01 — Gaussian conditioning

Independent blocks make conditioning leave the marginal unchanged

Carried forward S02 · joint Gaussian block model

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \right)$$

Illustrative independent case

$$X_1 \perp X_2 \quad \Rightarrow \quad \Sigma_{12} = 0, \quad \Sigma_{21} = 0$$

Joint Gaussian with zero cross blocks

$$X \sim \mathcal{N}\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & 0 \\ 0 & \Sigma_{22} \end{pmatrix} \right)$$

Independence factors the joint density

$$p(x_1, x_2) = p(x_1)p(x_2)$$

Carried forward S02 · conditional density rule

$$p(x_1 \mid x_2) = \frac{p(x_1, x_2)}{p(x_2)}$$

$$p(x_1 \mid x_2) = \frac{p(x_1)p(x_2)}{p(x_2)} = p(x_1)$$

- We will reproduce this conditioning pattern with an independent residual.

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STUDY NOTE

When the blocks are independent, the joint density factors, so conditioning on the second block leaves the first block's marginal law unchanged. We will reproduce this useful pattern for correlated blocks by replacing the first block with an independent residual.

YOUR NOTES

Recording L01-R01 — Gaussian conditioning

Seek independent coordinates that keep the observation

Carried forward S02 · joint Gaussian block model

$$X \sim \mathcal{N}\left(\begin{pmatrix}\mu_1\\ \mu_2\end{pmatrix}, \begin{pmatrix}\Sigma_{11} & \Sigma_{12}\\ \Sigma_{21} & \Sigma_{22}\end{pmatrix}\right)$$

$$Y = T(X-\mu) = \begin{pmatrix}Y_1\\ Y_2\end{pmatrix}$$

Search requirements	Why each matters
$Y_2 = X_2 - \mu_2$	Fixing x_2 fixes Y_2.
$Y_1 \perp X_2 \Leftrightarrow Y_1 \perp Y_2$	So Y_1 is not affect by fixing x_2.
T invertible	Recover X_1 from (Y_1, X_2).

- Once T is found, recover $X = T^{-1}Y + \mu$.
- Fix $Y_2 = x_2 - \mu_2$; independence leaves Y_1 unchanged.
- Use the first block to obtain $p(x_1 \mid x_2)$.

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STUDY NOTE

We seek transformed coordinates whose second block is the centered observation and whose first block is independent of it. The transformation must also be invertible so that the conditional law of the original first block can be recovered after fixing the observation.

YOUR NOTES

Recording L01-R01 — Gaussian conditioning

Independence removes cross blocks after taking expectations

Carried forward S07 · observation-preserving coordinates

$$Y = T(X-\mu) = \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix}, \quad Y_2 = X_2 - \mu_2$$

Center

$$E[Y] = T(E[X]-\mu) = 0, \quad D \triangleq \mathrm{Cov}(Y) = E[YY^T]$$

$$YY^T = \begin{pmatrix} Y_1Y_1^T & Y_1Y_2^T \\ Y_2Y_1^T & Y_2Y_2^T \end{pmatrix}$$

By assumption $Y_1 \perp Y_2 \quad \Rightarrow \quad E[Y_1Y_2^T] = E[Y_1]E[Y_2]^T = 0$

$$D = E[YY^T] = \begin{pmatrix} \mathrm{Cov}(Y_1) & 0 \\ 0 & \mathrm{Cov}(Y_2) \end{pmatrix}$$

- With x_2 given, the off-diagonal block is covariance with the observation.

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STUDY NOTE

For centered independent blocks, the expected cross products vanish, making their covariance block diagonal. This gives a concrete algebraic condition for the independent coordinates we seek.

YOUR NOTES

Recording L01-R01 — Gaussian conditioning

We need the transformed covariance to be block diagonal

Carried forward S03 · affine Gaussian covariance rule

$\mathrm{Cov}(TX) = T\Sigma T^T$

Carried over L01 S47 · Gaussian independence

(Y_1, Y_2) jointly Gaussian

$\mathrm{Cov}(Y_1, Y_2) = 0 \Rightarrow Y_1 \perp Y_2$

$$Y = T(X - \mu) \sim \mathcal{N}(0, T\Sigma T^T)$$

$$D = \mathrm{Cov}(Y) = T\Sigma T^T$$

$$Y_1 \perp Y_2 \Leftrightarrow \mathrm{Cov}(Y_1, Y_2) = 0$$

Choose T so that

$$T\Sigma T^T = \begin{pmatrix} \mathrm{Cov}(Y_1) & 0 \\ 0 & \mathrm{Cov}(Y_2) \end{pmatrix}$$

- Within observation-preserving maps, block-diagonal covariance tests independence from X_2.

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STUDY NOTE

The transformed vector is jointly Gaussian, so zero cross covariance between its blocks is enough to establish independence. We can therefore search for an observation-preserving transformation by making the off-diagonal covariance block zero.

YOUR NOTES

Recording L01-R01 — Gaussian conditioning

Keep the observation and subtract a linear prediction

Carried forward S07 · observation-preserving coordinates

$Y_2 = X_2 - \mu_2$

Choose the first coordinate by subtracting a linear prediction

$$R \triangleq X_1 - \mu_1 - B(X_2 - \mu_2), \quad Y_1 = R$$

Looking for $\mathrm{Cov}(R, X_2) = 0$

$$T = \begin{pmatrix} I_{d_1} & -B \\ 0 & I_{d_2} \end{pmatrix}, \quad Y = \begin{pmatrix} R \\ X_2 - \mu_2 \end{pmatrix} = T(X - \mu)$$

$$X_2 = \mu_2 + Y_2, \quad X_1 = \mu_1 + R + B(X_2 - \mu_2)$$

- Keep X_2 and recover X_1.
- A suitable B then answers the original conditional query.

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STUDY NOTE

Subtract a linear function of the observed block from the first block, while keeping the observation itself. This triangular transformation is invertible, so a coefficient that removes cross covariance will let us reconstruct the original conditional law.

YOUR NOTES

Recording L01-R01 — Gaussian conditioning

Cancel cross covariance to determine the prediction coefficient

Carried forward S10 · residual definition

$R = X_1 - \mu_1 - B(X_2 - \mu_2)$

$$\begin{aligned}\operatorname{Cov}(R, X_2) &= \operatorname{Cov}(X_1, X_2) - B\ \operatorname{Cov}(X_2, X_2) \\ &= \Sigma_{12} - B\Sigma_{22} = 0\end{aligned}$$

$$\Sigma_{12} - B\Sigma_{22} = 0 \quad \Leftrightarrow \quad B\Sigma_{22} = \Sigma_{12}$$

Since $\Sigma \succ 0$, its principal block Σ_{22} is invertible. Multiply on the right.

$$B\Sigma_{22}\Sigma_{22}^{-1} = \Sigma_{12}\Sigma_{22}^{-1}$$

$$B = \Sigma_{12}\Sigma_{22}^{-1}$$

- Zero cross covariance is in place.
- Joint Gaussianity will make R independent of X_2.

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STUDY NOTE

The residual's cross covariance with the observation is the original cross covariance minus the coefficient times the observation covariance. Setting it to zero determines the coefficient; the positive definite observation block makes the required inverse available.

YOUR NOTES

Recording L01-R01 — Gaussian conditioning

The chosen subtraction leaves an independent Gaussian residual

Carried forward S10 · residual definition; S11 · zero cross-covariance

$$R = (X_1 - \mu_1) - B(X_2 - \mu_2)$$

$$\mathrm{Cov}(R, X_2) = 0$$

Use the block means

$$E[R] = E[X_1] - \mu_1 - B(E[X_2] - \mu_2) = 0$$

$$(R, X_2) \text{ jointly Gaussian}$$

$$R \perp X_2$$

- Conditioning on X_2 leaves the residual law unchanged.

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STUDY NOTE

The chosen residual has mean zero and is jointly Gaussian with the observation. Its zero cross covariance therefore gives independence, so conditioning on the observed block will not change the residual's distribution.

YOUR NOTES

Recording L01-R01 — Gaussian conditioning

Residual covariance starts with an expected outer product

Carried forward S10 · residual definition; S11 · coefficient B and zero cross-covariance; S12 · zero residual mean

$E[R] = 0$

$B = \Sigma_{12}\Sigma_{22}^{-1}$

$R = (X_1 - \mu_1) - B(X_2 - \mu_2)$

$\mathrm{Cov}(R, X_2) = 0$

$$
\begin{aligned}
S &\triangleq E[RR^T] \\
&= E\Big[((X_1 - \mu_1) - B(X_2 - \mu_2))\big((X_1 - \mu_1)^T - (X_2 - \mu_2)^T B^T\big)\Big] \\
&= E\big[(X_1 - \mu_1)(X_1 - \mu_1)^T\big] - E\big[(X_1 - \mu_1)(X_2 - \mu_2)^T\big]B^T \\
&\quad - BE\big[(X_2 - \mu_2)(X_1 - \mu_1)^T\big] + BE\big[(X_2 - \mu_2)(X_2 - \mu_2)^T\big]B^T
\end{aligned}
$$

- The four expectations make S computable from the supplied covariance blocks.

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STUDY NOTE

Because the residual is centered, its covariance is the expectation of its outer product. Expanding that product into four terms expresses the remaining uncertainty entirely through known blocks of the joint covariance.

YOUR NOTES

Recording L01-R01 — Gaussian conditioning

The four terms reduce to the residual covariance

Carried forward S11 · coefficient B; S13 · four-term covariance expansion

$B = \Sigma_{12}\Sigma_{22}^{-1}$

$$S = E\left[(X_1-\mu_1)(X_1-\mu_1)^T\right] - E\left[(X_1-\mu_1)(X_2-\mu_2)^T\right]B^T$$
$$-BE\left[(X_2-\mu_2)(X_1-\mu_1)^T\right] + BE\left[(X_2-\mu_2)(X_2-\mu_2)^T\right]B^T$$

Simplify S

$$S = \Sigma_{11} - 2\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$
$$+\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$
$$= \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

Term calculations

$$B\Sigma_{21} = \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$
$$\Sigma_{12}B^T = \Sigma_{12}\left(\Sigma_{22}^{-1}\right)^T\Sigma_{12}^T \qquad \Sigma_{22}^T = \Sigma_{22},\ \Sigma_{12}^T = \Sigma_{21}$$
$$= \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$
$$B\Sigma_{22}B^T = \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{22}B^T$$
$$= \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

- S is the residual spread left after subtracting the linear prediction.

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STUDY NOTE

Substituting the covariance blocks and the chosen coefficient makes three adjustment terms combine into one subtraction. The resulting Schur complement is the covariance of the independent residual, hence the spread of the conditional law.

YOUR NOTES

Recording L01-R01 — Gaussian conditioning

The residual coordinates give the block-diagonal target

Carried forward S10 · residual coordinates; S12 · independence; S14 · residual covariance

$$Y = \begin{pmatrix} R \\ X_2 - \mu_2 \end{pmatrix} = T(X - \mu)$$

$$D \triangleq \mathrm{Cov}(Y) = T\Sigma T^T$$

$$\mathrm{Cov}(R) = S, \quad R \perp X_2$$

Independence removes the cross blocks; centering preserves covariance

$$\mathrm{Cov}(R, X_2 - \mu_2) = \mathrm{Cov}(R, X_2) = 0$$

$$\mathrm{Cov}(X_2 - \mu_2) = \Sigma_{22}$$

$$D = \begin{pmatrix} S & 0 \\ 0 & \Sigma_{22} \end{pmatrix}$$

- Residual coordinates make R independent of X_2.
- The observed block retains covariance Σ_{22}.

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STUDY NOTE

The residual and centered observation now have block-diagonal covariance, with the residual covariance given by the Schur complement. This confirms that the transformed coordinates satisfy the independence condition used to derive the conditional law.

YOUR NOTES

Recording L01-R01 — Gaussian conditioning

Conditioning cannot increase variance in any direction

Carried forward S14 · Schur complement

$S = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$

Since $\Sigma \succ 0$, also $\Sigma_{22} \succ 0$. Using $B = \Sigma_{12}\Sigma_{22}^{-1}$ and symmetry of Σ:

$$\Sigma_{11} - S = \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} = B\Sigma_{22}B^T$$

Fix x_2: only the residual varies

$$\mathrm{Cov}(X_1 \mid X_2 = x_2) = \mathrm{Cov}(R) = S$$

For any $v \in \mathbb{R}^{d_1}$

$$\begin{aligned} v^T(\Sigma_{11} - S)v &= v^T B\Sigma_{22}B^T v \\ &= (B^T v)^T \Sigma_{22}(B^T v) \\ &\geq 0 \end{aligned}$$

Variance cannot increase

$$S \preceq \Sigma_{11}$$

$$\begin{aligned} \mathrm{Var}(v^T X_1 \mid X_2 = x_2) &= v^T S v \\ &\leq v^T \Sigma_{11} v \\ &= \mathrm{Var}(v^T X_1) \end{aligned}$$

For $B = 0$, $S = \Sigma_{11}$.

- Conditioning reduces or preserves variance in every direction.
- In this Gaussian model, S does not depend on the observed value x_2.

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STUDY NOTE

The difference between marginal and conditional covariance is a positive semidefinite matrix. For any linear direction, observing the second block therefore reduces or preserves variance; in this Gaussian model the amount of residual variance does not depend on the observed value.

YOUR NOTES

Recording L01-R01 — Gaussian conditioning

The two queries have explicit Gaussian laws

$$p(x_1) = \mathcal{N}(x_1 \mid \mu_1, \Sigma_{11})$$

$$p(x_1 \mid x_2) = \mathcal{N}(x_1 \mid m(x_2), S)$$

Conditional parameters

$$m(x_2) = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$

$$S = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

TAKE HOME

- Select a block to obtain its marginal law.
- Fix x_2 and shift the independent residual to obtain the conditional law.

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STUDY NOTE

Selecting a block gives its marginal Gaussian law. For conditioning, fix the observed block and shift the independent Gaussian residual: the conditional mean changes with the observation, while its covariance is the Schur complement.

YOUR NOTES